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MILITARY SPECIFICATION

BOILERS, STEAM, HIGH PRESSURE, AUXILIARY FIRE TUBE TYPE NAVAL SHIPBOARD

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fuel fired horizontal straight fire tube type boilers of 2 or 4 pass, single uptake design with burner, all necessary appurtenances, controls and auxiliary equipment mounted as an integral unit on a steel base in order to make a complete, self-contained steam generating unit.

1.2 Classification. The boilers shall be of the following classes and sizes, as specified (see 6.2.1):

Class I - Carbon steel construction.
Class II - Nonmagnetic construction.

Size	Capacity (see 6.6)	
	Rate (pounds of steam per hour)	Overload (pounds of steam per hour)
1	500	----
2	800	----
3	1200	1440
4	1500	1800
5	2500	3000
6	4000	4800
7	5000	6000

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

- TT-P-28 - Paint, Aluminum, Heat Resisting (1200°F).
- PPP-B-601 - Boxes, Wood, Cleated-Plywood.
- PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
- PPP-B-636 - Boxes, Shipping, Fiberboard.
- PPP-B-1055 - Barrier Material, Waterproofed, Flexible.
- PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

- MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
- MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
- MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
- MIL-R-2033 - Relays - Power, Electrical Specialty, Induction Type, Naval Shipboard Use.
- MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
- MIL-C-2212 - Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
- MIL-R-2729 - Regulator Sets, Voltage, A.C. Generator, Naval Shipboard Use.
- MIL-G-3124 - Generator, Alternating Current, 60-Cycle (Naval Shipboard Use).
- MIL-B-3180 - Boilers and Related Equipment, Packaging of.
- MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
- MIL-C-5015 - Connectors, Electrical, Circular Threaded, General Specification for.
- MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.

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MILITARY (Continued)

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-S-16165 - Shielding Harnesses, Shielding Items and Shielding Enclosures for Use in the Reduction of Interference from Engine Electrical Systems.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-W-16878 - Wire, Electrical, Insulated, General Specification for.
MIL-F-16884 - Fuel, Naval Distillate.
MIL-M-17059 - Motors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use).
MIL-M-17060 - Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Including Repair Parts, Tools, Accessories and Instruments); Packaging of.
MIL-T-17370 - Tube and Flue Cleaner, Boiler, Internal, Air-Motor Driven.
MIL-M-17413 - Motors, Direct Current, Integral H.P., Naval Shipboard.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Provisioned Items (Repair Parts): Packaging of.
MIL-M-17556 - Motor, Direct-Current, Fractional HP (Shipboard Use).
MIL-P-17945 - Pumps, Centrifugal, Peripheral Type, Miscellaneous Naval Shipboard Use.
MIL-F-19004 - Fans, Centrifugal, Fixed and Portable Ventilation, Naval Shipboard.
MIL-I-24137 - Iron Castings, Nodular Graphitic (Ductile Iron) and Nodular Graphitic (Corrosion Resisting, Austenitic, Low Magnetic Permeability) (For Shipboard Application).
MIL-C-24643 - Cable and Cord, Electrical, Low Smoke, For Shipboard Use, General Specification for.
MIL-C-24643/23 - Cable, Electrical, 300 Volts, Type LSTTSU (Including Variation LSTTSA).

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STANDARDS

MILITARY

MIL-STD-22	- Welded Joint Design.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278	- Welding and Casting Standard.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-777	- Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.

2.1.2 Other Government drawing. The following other Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
810-1385850 - Piping, Instrument, Pressure for All Services.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code, Section I

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging.
(DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Material. Cast iron shall not be used in any part of the boiler or in any part of the boiler auxiliaries. Nodular iron conforming to MIL-I-24137 may be used in such parts of the boiler auxiliaries where no other substitute for cast iron is practicable.

3.2.1 Class I - carbon steel construction. Except as otherwise specified herein, material used in construction of class I boilers shall conform to the ASME Boiler and Pressure Vessel Code. Material for tubes shall be equal to or better than SA-178, grade C. Material for shell and tube sheets shall be equal to or better than ASME material SA-285, grade C, firebox quality, with respect to maximum allowable stress taken at 500 degrees Fahrenheit (°F) and weldability.

3.2.2 Class II - nonmagnetic construction. Boilers specified herein are to be installed on ships whose magnetic signature must be extremely low. Therefore, great importance is attached to use of nonmagnetic materials wherever possible without involving extensive development or affecting the design characteristics. Unless otherwise specified herein, the ASME Boiler and Pressure Vessel Code shall apply.

3.2.2.1 Nonmagnetic materials. Components used in boiler assembly, other than the boiler shell assembly, and pressure vessel parts (for example, handhole plates, pipe and nipples attached to boiler shell) shall be manufactured from nonmagnetic materials. Except as otherwise specified herein, material used in the construction of equipment shall be considered nonmagnetic if it has a permeability of less than 1.08 before fabrication and not greater than 2.0 after fabrication.

3.2.2.2 Carbon steel. Boiler shell assembly, tube sheets, and the tube assembly shall be carbon steel as specified in 3.2.1.

3.2.2.3 Electrical equipment. Electrical equipment for class II boilers shall be as follows:

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(a) Each item of electrical equipment shall be fabricated of nonmagnetic material, except for the following parts:

(1) Parts required in the magnetic circuit of the particular item of equipment.
(2) Structural parts of motors, rotary converters, contactors, relays, thermostats, and pressure switches that would entail extensive modification and redesign to permit substitution of nonmagnetic material.

(b) Electric controllers, solenoids, and switches shall have enclosures and panels of nonmagnetic metals.
(c) Stray magnetic fields caused by electrical circuits shall be kept to a minimum by making the area of electric current loops as small as possible and arranging adjacent loops so that the fields will be in opposition. Current loops shall enclose no magnetic material unless essential for proper performance of equipment.
(d) In alternating current (ac) motors, all connections between stator winding connections shall be made as short as possible. The power leads to the stator winding shall be run through the same opening in the motor frame.

3.3 Construction. Except as otherwise specified herein, the boiler shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code.

3.3.1 General configuration. The boiler shall be fuel fired horizontal straight fire tube type of the 2 or 4 pass design. The boiler shall be either right or left hand as specified (see 6.2.1). The hand of the boiler shall be determined by the position of the control panel box when facing the firing end of the boiler.

3.3.2 Design and operating characteristics. The boiler shall have the design and operating characteristics specified in table I.

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TABLE I. Design and operating characteristics.

Characteristic	Sizes 1 and 2	Sizes 3 through 7
Operating pressure (see 6.7)	35 lb/in^2	125 lb/in^2
Design pressure	100 lb/in^2	150 lb/in^2
Safety valve pressure setting		
First valve	45 lb/in^2	145 lb/in^2
Second valve, if used	------	150 lb/in^2
Exit gas temperature at overload capacity (see 4.5.1.1)	550°F	650°F
Capacity, rated; lb/hr	As specified (see 1.2, 6.2.1 and 6.6)	
Efficiency at rated capacity (see 4.5.1.1)	78 percent minimum	
Fuel burned	Fuel, naval distillate (NATO Symbol F-76), in accordance with MIL-F-16884 or JP-5 specified in accordance with MIL-T-5624 as specified (see 3.5.1 and 6.2.1)	

3.3.3 Electrical characteristics. The boiler and auxiliaries shall operate on the electric power of the rating specified (see 6.2.1).

3.3.4 Shock resistance. Unless otherwise specified (see 6.2.1), the first article unit shall be tested for shock (see 4.5.2).

3.3.5 Hydrostatic test. The boiler shall be hydrostatically tested (see 4.5.3).

3.3.6 Method of operation. Method of operation shall be as specified in 3.3.6.1 through 3.3.6.3.2.

3.3.6.1 Ignition. Boilers shall be equipped for electric spark ignition, automatic or manual, as applicable.

3.3.6.2 Feedwater regulation. Boilers shall be equipped for automatic feedwater regulation (see 3.6.2).

3.3.6.3 Burner operation.

3.3.6.3.1 Boiler sizes 1 and 2. Boilers, sizes 1 and 2, shall be equipped for automatic ON and OFF type of burner operation.

3.3.6.3.2 Boiler sizes 3 through 7. Boilers, sizes 3 through 7, shall be equipped for one of the following methods of burner operation, as specified (see 6.2.1):

(a) Manual operation - A unit manually operated through turndown range and manually restarted following shutdown.
(b) Automatic operation - A unit that will automatically modulate through turndown range with an automatic restart feature following shutdown.

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(c) Semiautomatic operation - A unit that will automatically modulate through turndown range without automatic restart feature. The unit must be manually restarted following shutdown.

3.3.7 Boiler internals. The boiler shall be equipped with the following internal fittings:

(a) Dry pipe.
(b) Surface blow pipe.
(c) Feed pipe.
(d) Baffle plates, as required.

3.3.8 Water gauges. Water gauges, separately mounted, with valves shall be located on the front end of the boiler as close to the vertical center line of the boiler as is practicable. They shall be the reflex type with 10-inch minimum visible range. The center of the gauge glass shall correspond to the normal water level of the boiler.

3.3.9 Valves. Each valve, including the safety valves, shall conform to the requirements of the ASME Boiler and Pressure Vessel Code, except for class I boilers, the safety valves shall be cast or forged steel and all other valves shall be bronze or cast or forged steel; for class II boilers, all valves shall use nonmagnetic material.

3.3.10 Surface blow valve. The surface blow valve shall be located so the operator can observe the water level in the water gauge while operating the valve.

3.3.11 Safety valve hand easing gear. If the safety valve cannot be reached from the boiler operating level, a safety valve hand easing gear shall be installed to provide a simple, effective mechanical means for raising the valve so that operating controls are at or adjacent to firing front of the boiler (see 4.5.4).

3.3.12 Steam blanket connection. A connection and valve shall be provided in a location that will admit blanketing steam to the boiler water sides.

3.3.13 Boiler base. Boiler and all auxiliaries shall be firmly supported on a base. Means shall be provided for connecting the base to the ship's deck or foundation.

3.3.14 Supply air and smoke pipe connections. Provisions shall be made for connecting an air supply duct from the weather to forced draft blower and for connecting a smoke pipe to the boiler smoke outlet.

3.3.15 Sample cooler. A sample cooler shall be provided to cool water samples drained from the boiler for testing.

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3.3.16 Insulation and lagging. The boiler shall be equipped with insulation and lagging consisting of a metal jacket over the boiler shell insulation.

3.3.16.1 Paint. Exterior surfaces of boiler shell assembly that are subject to corrosion shall be given two coats of heat-resistant paint conforming to TT-P-28 before application of insulation and lagging.

3.3.16.2 Surface temperature. Parts of the boiler assembly shall be insulated so that at overload operation the average surface temperature of the casing shall not exceed 135°F, (front, rear and each side considered separately), based upon an ambient air temperature of 100°F, and a velocity of 0.5 feet per second taken 1 foot away from the casing. The average boiler casing temperature shall not exceed 135°F (see 4.5.1.1).

3.3.17 Refractory. Refractory walls and liners of brick construction in furnace and dryback assembly application are not acceptable. Castable or plastic refractory suitable for temperature of the environment shall be used. The refractory shall be suited for the particular application and shall be reinforced by anchor bolts, strips, or nichrome wire, as necessary.

3.3.18 Shell assembly. The boiler shell assembly shall be of welded construction. The welding shall be in accordance with MIL-STD-278, class A-1 pressure vessels.

3.3.18.1 Tube sheet. The tube sheet shall be of the flat or flat disc flanged construction in accordance with the ASME Boiler and Pressure Vessel Code.

3.3.18.2 Welding. Longitudinal seams in the boiler shell and circumferential joints between the tube sheets and the shell cylinder and between the tube sheets and the combustion chamber shall be welded from inside and outside and shall conform to the butt joints B2S.1, B2V.1 or B2V.3 of MIL-STD-22, except for the following:

(a) Joints that cannot be welded on inside shall comply with B1V.1 of MIL-STD-22.
(b) When location of the combustion chamber in the shell renders flanging of the tube sheet for the butt welded joint between the tube sheets and the combustion chamber impracticable, a single J tee joint, fillet reinforced weld shall be used. This joint shall be similar to joint number T1J.1, except that the root gap shall not exceed 1/8 inch.

3.3.18.3 Normal water level stamping. The words "Normal Water Level" or initials "NWL" shall be stamped on the front of the boiler shell to indicate the location of the normal water level.

3.3.19 Tube assembly. Tube assembly shall be as specified in 3.3.19.1 through 3.3.19.2.

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3.3.19.1 Tube sheet holes. The finished diameters of tube holes in tube sheets shall not exceed the following diameters:

Size tube (inches)	Tube hole diameter (inches)
1	1.02
1-1/4	1.28
1-1/2	1.53
1-3/4	1.78
2	2.03
2-1/4	2.28
2-1/2	2.53

3.3.19.1.1 Edges. The edges of the tube holes shall be dressed free of all projecting metal and sharp edges.

3.3.19.2 Tubes. Tubes shall be connected to both front and rear tube sheets by rolling, beading, and welding around the edge of the bead. Tubes may have extended surface (fin tubes) or gas side turbulators.

3.3.20 Auxiliaries, accessories and controls. The boiler shall be equipped with all auxiliaries, accessories, and controls to make a self-contained and complete steam-generating unit when connected with electric power, fuel oil, and feedwater supplies and return lines. No economizers, air heaters or superheaters shall be fitted to the boiler. The auxiliaries, accessories and controls shall include the following:

(a) Burner (see 3.6.1).
(b) Automatic ignition and combustion control system (see 3.3.6.1 and 3.6.6).
(c) Flame failure safety control (see 3.6.6 and 3.6.7).
(d) Fuel system (see 3.6.3).
(e) Forced draft blower and air regulating mechanism (see 3.6.4).
(f) Feedwater system, including feedwater treatment tank (see 3.6.5).
(g) Automatic feedwater regulator and low water cutoff control (see 3.6.2 and 3.6.7).
(h) Motors and controllers, as required by the driven auxiliaries (see 3.6.7 and 3.6.8).
(i) Rotary inverter to supply ac for automatic ignition when direct current (dc) power supply is specified (see 3.3.3, 3.6.7 and 6.2.1).
(j) Safety valves, as required (see 3.3.2 and 3.3.9).
(k) Safety valve hand easing gear, if required (see 3.3.11).
(l) Water gauges (see 3.3.8).
(m) Sample cooler (see 3.3.15).
(n) Steam pressure gauge (panel mounted flush), gauge piping to conform to Drawing 810-1385850.
(o) Temperature indicators: Combustion gas outlet thermometer and feedwater thermometer.

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(p) Combustion chamber observation window. Front and rear with scavenger air attached.
(q) Steam stop valve (see 3.3.9).
(r) Feedwater stop valve (see 3.3.9).
(s) Feedwater check valve (see 3.3.9).
(t) Bottom blow valves (see 3.3.9).
(u) Surface blow valves (see 3.3.9 and 3.3.10).
(v) Air vent valve (see 3.3.9).
(w) Steam blanket valve (see 3.3.9 and 3.3.12).

3.4 Maintainability. Internal parts shall be constructed to permit easy disassembly and reassembly with standard tools and to prevent as is practical the incorrect reassembly of parts. Positioning and alignment of all parts in assembly shall employ positive means so that correct reassembly is repeatedly assured. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.4.1 Manholes and handholes. The boiler shell shall have a manhole, as required by the ASME Boiler and Pressure Vessel Code, and a minimum of six handholes located in order to permit easy inspection and access to all parts of the shell assembly for flushing of the water sides.

3.4.2 Tube configuration. The configuration of all boiler tubes shall permit mechanical cleaning of firesides using air driven tube cleaning equipment conforming to MIL-T-17370.

3.5 Performance characteristics. Performance characteristics shall be as specified in 3.5.1 through 3.5.3.

3.5.1 Fuel. The boiler shall burn and utilize Naval distillate fuel (NATO Symbol F-76) conforming to MIL-F-16884 and JP-5 conforming to MIL-T-5624. For heat transfer calculation purposes, the analysis and higher heating value of the fuel shall be considered as:

Carbon	85.8	percent
Hydrogen	13.0	percent
Sulphur	1.0	percent
Nitrogen	0.10	percent
Oxygen and undetermined	0.10	percent
British thermal units (Btu)/pound	19,300	

3.5.2 Steam moisture content. When the boiler is operated in conditions specified in 3.7, the moisture content in the steam shall not exceed 1 percent at all evaporating rates, including overload operation, with the maximum chloride and total solids concentration in boiler water of 550 and 1175 parts per million (p/m) respectively.

3.5.3 Overload operation. Boilers, sizes 3 through 7, shall overload operate at least 120 percent of their rated capacity.

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3.6 Details of components. Components shall be as specified in 3.6.1 through 3.6.1.2.1.

3.6.1 Burner. The burner shall be a mechanical pressure atomizing type. The burner shall have smokeless operation over the entire boiler operating range when burning fuel of the required type (see 3.5.1).

3.6.1.1 Boiler sizes 1 and 2. For boilers, sizes 1 and 2, the burner shall be arranged for a constant rate fully automatic operation by means of ON and OFF cycling (see 3.6.6.1).

3.6.1.2 Boiler sizes 3 through 7. For boilers, sizes 3 through 7, the burner shall be a variable capacity wide range type able to operate at all firing rates corresponding to boiler steaming loads from the overload capacity to 20 percent of the overload capacity without change of the sprayer plates or burner nozzle tips (see 3.6.6.2).

3.6.1.2.1 Manual or semiautomatic operation. When manual or semi-automatic operation is specified (see 3.2.6), the electric spark ignition and the flame failure control (see 3.6.7) shall be simultaneously energized by manually operated push button.

3.6.2 Feedwater regulator and low water cutoff control. The automatic feedwater regulator and low water cutoff control shall be of an electrode type designed for flange mounting on the top of the boiler shell at a location equidistant from the tube sheets. The control shall be arranged to function without short cycling when water in the boiler is shifting because of conditions specified in 3.7. The control shall operate the feedwater pump between the low and the high water level limits and shall stop boiler operation when the lowest permissible water level is reached. After this emergency cutoff, the boiler shall be restarted only by manually operating the pushbutton master switch on the control panel. When the boiler is operated in conditions specified in 3.7, the water level shall hold within plus or minus 4 inches at all steaming rates and during ON-OFF operation.

3.6.2.1 Feedwater pump. Provisions shall be made for manual operation of the feedwater pump in case automatic operation is not desired or prevented.

3.6.3 Fuel system. The fuel system shall include a fuel oil strainer, fuel pump, fuel filter (with nondisposable element), relief valve, fuel solenoid valve (dual for class 3 through 7), fuel oil return control valve and interconnecting fittings. Means shall be provided for disconnecting the following components from the fuel system for maintenance: burner, fuel oil pump, fuel oil solenoid valves, pressure regulator, and the fuel oil return control valve.

3.6.3.1 Fuel system piping and fittings. The fuel oil system piping and fittings shall be in accordance with MIL-STD-777 with the following exceptions:

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(a) Piping threads may be utilized. However, their use shall be minimized and when used, they shall be seal brazed.
(b) Stainless steel tubing shall be used from the solenoid valve or valves to the burner, and from the burner to the fuel oil return control valve.

3.6.3.2 Strainers and filters. Fuel strainers and filters shall be either the duplex type or the simplex-self-cleaning kind so that no interruption in flow can occur when cleaning.

3.6.3.3 Fuel pump. The fuel pump shall be self-priming and shall have a minimum suction lift of 12 feet. The pump shall supply all requisite fuel and shall have characteristics for operation with the burner furnished. The pump shall have a direct coupled transmission and separate motor drive (motor shall have no other loading).

3.6.3.4 Operation. When automatic or semiautomatic operation is specified, the system shall be arranged to permit manual regulation throughout the entire firing range of the burner, in addition to automatic regulation.

3.6.4 Forced draft blower. The forced draft blower shall operate with a negative pressure of 3/10 inch of water at the suction end, imposed by resistance in the air supply duct between the blower air intake and the weather, and shall supply all requisite air to the boiler at a pressure at least 0.5 and 1.0 inch of water for boilers, sizes 1 and 2, and boilers, sizes 3 through 7, respectively, in excess of the maximum pressure required to overcome all resistances of the burner, and the boiler between the air intake of the air regulating control and the exhaust gas outlet of the boiler. The blower wheel (fan) shall be in satisfactory static and dynamic balance. The interior of the blower casing shall be smooth and free from obstructions such as bolt heads, nuts, and pipe fittings. The following features shall apply:

(a) The wheel shall be keyed on the shaft.
(b) Provision shall be made for two set screws in the hub located 90 degrees apart.
(c) One set screw shall be located over the key.
(d) One set screw shall be recessed in the shaft for positioning axially.
(e) Frictionless type of bearings shall be employed and the shaft shall be provided with a shoulder or collar to prevent axial thrust.
(f) The blower shall be direct driven by an electric motor that shall have no other loading. The maximum usable pressure of the fan shall be determined by the total pressure-volume curve in accordance with the procedures specified in MIL-F-19004.

3.6.4.1 Air regulating control. An air regulating control shall be provided for controlling air supply to the burner in correct proportion to its firing rates. The control shall be interlocked with the fuel oil return control valve in the fuel system. The combustion control system shall provide

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simultaneous manual regulation of the fuel and air supplies to the burner when boiler is steaming at variable loads. Means shall be provided in the combustion control system for manual adjustment of fuel air ratio as necessary for smokeless operation of the burner throughout its entire firing range.

3.6.5 Feedwater system. The feedwater system shall include a water strainer, feed pump, relief valve, pressure gauge, feed treatment tank, feedwater stop and check valves and necessary connecting piping, valves, and fittings.

3.6.5.1 Feedwater pump. The boiler feedwater pump shall be of a centrifugal or a turbine type that shall feed requisite water at 190°F to the boiler steaming at maximum capacity with a pressure of 175 pounds per square inch (lb/in^2). The pump shall be self-priming and shall have a minimum suction lift of 12 feet. Pump discharge head shall have a practical margin when injecting water into the boiler through a feed line containing a feed water regulator control valve, and stop and check valves. Design and construction shall be in accordance with MIL-P-17945. The pump shall have a direct coupled drive. The motor drive shall have no other loading.

3.6.5.1.1 Feedwater pump time delay relay. The feedwater pump shall have a time delay relay adjustable from 0 to 20 seconds to minimize on-off cycling of the pump motor because of wave action of the boiler water.

3.6.5.2 Feedwater treatment tank. The feedwater treatment tank shall be installed in a bypass branch in the feedwater system located between the feed pump relief valve and the feedwater stop and check valves. The tank shall be valved so that feedwater can be passed through the main feedwater line or bypassed through the branch containing the tank. The tank shall be equipped with a funnel for adding chemicals, and check valve between the funnel and tank, a vent valve, and a drain valve.

3.6.6 Automatic ignition and combustion control system operation. The automatic ignition and combustion control system shall be arranged for the following operation:

3.6.6.1 Boiler sizes 1 and 2. Boiler sizes 1 and 2 shall be as specified in 3.6.6.1.1 and 3.6.6.1.2.

3.6.6.1.1 Operation. The automatic ignition and combustion control system shall have an adjustable timer and be arranged for the following sequence of operation upon the initial start and all subsequent automatic restarts during on and off cycling:

(a) A pre-purge period to provide at least five volumetric air changes of the combined volume of boiler furnace, uptake and smoke pipe.
(b) At the end of the pre-purge period the ignition circuit shall be energized with the fuel valve circuit.

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(c) If flame is detected, the ignition circuit shall be de-energized 20 seconds after opening of the fuel solenoid valve.
(d) If flame is not detected, the blower shall operate through a post-purge period of at least 15 seconds following shutoff of the fuel solenoid valve and securing of the fuel oil pump.

3.6.6.1.2 Ignition failure. The automatic ignition and combustion control system shall be arranged to shut off the fuel solenoid valve 2 to 4 seconds after opening in case of ignition failure. Following this shutoff of the fuel solenoid valve and securing of the fuel oil pump, the blower motor shall continue operation for at least 15 seconds to purge the combustion chamber. After completion of the purging period, a safety shutdown of burner operation shall take place. Upon such safety shutdown, the burner shall be restarted only by manually operating the pushbutton master switch on the control panel.

3.6.6.2 Boiler sizes 3 through 7. Boiler sizes 3 through 7 shall be as specified in 3.6.6.2.1 through 3.6.6.2.5.

3.6.6.2.1 Operation. The automatic ignition and combustion control system shall automatically modulate the burner without on and off cycling or change of sprayer plates through its entire firing range when the boiler is steaming at variable loads between the overload capacity and 20 percent of the overload capacity, and to automatically cycle the burner on and off when the boiler is steaming at loads under 20 percent of the overload capacity. The programmer shall not have manual override capability which will preempt the normal timing cycle/sequences. The automatic ignition and combustion control system shall be arranged for the following sequence of operation upon the initial start and all subsequent automatic restarts during the on and off cycling:

(a) A pre-purge period to provide at least five volumetric air changes of the combined volume of boiler furnace, uptake and smoke pipe. This period shall begin with the burner in the low fire position, move to the high fire position for at least 15 seconds and then move back to the low fire position for the balance of the pre-purge period. Boiler draft air shall be proven (running interlock) when the burner reaches the high fire position during the pre-purge period and remain proven through the run period.
(b) At the end of the pre-purge period, if the flame scanner does not detect flame in the boiler and if the burner is in the low fire position, the ignition circuit shall be energized. The fuel solenoid valves shall be energized with ignition circuit. If flame is detected, the ignition circuit shall be de-energized 20 seconds after the opening of the fuel solenoid valves.
(c) After flame is proven, the burner shall move to the position demanded by the modulating control.

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(d) The fuel solenoid valve shall be instantly de-energized and the fuel oil pump secured when operating controls call for the burner shutdown.
(e) Post-purge period shall be at least 15 seconds upon each shutdown of the fuel solenoid valves. During the post-purge period the burner shall move to the high fire position to be ready for the next operating cycle.

3.6.6.2.2 Ignition failure. The combustion control system shall shut off the fuel solenoid valves in 2 to 4 seconds after opening in the case of ignition failure. The fuel oil pump shall also be secured if ignition failure occurs. The blower motor shall continue operation for at least 15 seconds to purge the combustion chamber. During this period, the burner shall move to the high fire position. After completion of the purging period, a safety shutdown of burner operation shall take place. Upon such safety shutdown, the burner shall be restarted only by manually operating the reset button on the control panel.

3.6.6.2.3 Semiautomatic operation. When semiautomatic operation is specified, the automatic modulating feature shall automatically regulate the burner throughout its entire firing range.

3.6.6.2.4 Automatic operation. When automatic operation is specified, following shut-off because of flame failure, the blower motor shall continue operation for at least 15 seconds. During this period, the burner shall move to high fire position. After completion of the purging period, the burner shall shutdown on safety. Following safety shutdown, the burner shall be restarted only by manually operating the reset button. Following shutdown for any reason, provision shall be made to ensure that the restart can only be made at the low fire position.

3.6.6.2.5 Timing periods. Timing periods of the automatic ignition and combustion control system shall not vary because of ambient temperature variation between 32 and 120°F or because of supply voltage variation of plus 10 minus 15 percent of the nominal voltage. In case of a failure of supply voltage, the programming control shall automatically recycle when power is restored, following the prescribed sequence when protective limiting devices have been reset.

3.6.7 Automatic ignition and combustion control system components. The control circuit, including motor controller control relay circuits but excluding motor supply voltage, shall operate with 110 volts electric supply. A transformer shall be furnished for the control circuit in accordance with available service current (see 3.3.3). Ambient temperature shall be the prime consideration governing location of the control circuit transformer on the boiler assembly. One of the control system transformer secondary leads shall be grounded. Furthermore, the control system shall be constructed to ensure solenoid actuated fuel oil shutoff valves will close if a ground fault occurs in the control system. A rotary inverter shall be provided, as necessary, in accordance with available service current (see 3.3.3). The combustion control system and the automatic ignition circuit shall include the following:

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(a) Relays (as necessary) (see 3.6.7.1(d)).
(b) Switches (as necessary) (see 3.6.7.1(e) and (f)).
(c) Solenoid valves (as necessary) (see 3.6.7.1(g)).
(d) Fuel oil return control valve (see 3.6.7.1(h)).
(e) Combustion safeguard (see 3.6.7.1(i)).
(f) Control circuit transformer (see 3.6.7.1(j)).
(g) Ignition transformer (see 3.6.7.1(k)).
(h) Modulating motor (see 3.6.7.1(l)).
(i) Rotary inverter, if used (see 3.6.7.1(m)).
(j) Water level control (see 3.6.7.1(n)).
(k) Motor controllers (see 3.6.7.1(o)).
(l) Control enclosure (see 3.6.7.1(q)).
(m) Modulating pressure switch (boilers sizes 3 through 7).
(n) Pressure limit switch (steam).
(o) Time delay relay (see 3.6.5.1.1).
(p) Sequencing timer or programmer.
(q) Ignition electrodes.
(r) Fuses (as necessary).
(s) Indicator lights (see 3.6.7.1(r)).
(t) Audible alarm (see 3.6.7.1(s)).
(u) Boiler Master Automatic/Manual Control Station (see 3.6.7.1(t)).

3.6.7.1 Detail of components. Each control component, safety device, and circuit or interconnecting wiring shall be as simple as possible and shall be in accordance with MIL-E-917 and the following:

(a) Mercury tube actuated controls are not acceptable.
(b) Each component designed for HI-shock shall withstand type I vibration tests as specified in MIL-STD-167-1 without mechanical damages, contact chatter or other malfunctions.
(c) Electrical wire and cable shall conform to MIL-C-24643, MIL-C-224643/23, MIL-C-5015 and MIL-W-16878.
(d) Relays. Each relay, except time delay relay shall be high impact, shockproof units in accordance with MIL-R-2033 and the following:

Type - 60 hertz (Hz) or dc.
Tripping characteristics - As required.
Voltage - As required.
Duty - Continuous or intermittent as required.
Ambient temperature - 50 degrees Celsius (°C).

(e) Switches. Limit switches shall have a remote location mounted near the parent machine. Switches and push buttons shall be HI-shock design and shall conform to MIL-C-2212 and the following:

Voltage rating - As required.
Location - Local (mounted on operating panel).
Type - Push button or rotary.
Ambient temperature - 50°C.
Enclosure - Dripproof.

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(f) Pressure switches. Each pressure switch except the modulating pressure control switch shall be HI-shock design and shall conform to MIL-C-2212 and the following:

Voltage rating - 110 volts ac.
Location - Remote.
Type - Pressure actuated.
Ambient temperature - As required (50°C minimum).
Enclosure - Dripproof.

(1) A high steam pressure cutout switch that will secure the boiler in the event of failure of the pressure limit control shall be provided. This switch shall be mounted separately from the modulating pressure control switch. Once this switch is actuated, the boiler shall be restarted only by manually operating the push button master switch on the control panel. The high pressure cutout switch shall operate satisfactorily (see 4.5.1).

(2) An air safety switch shall be installed adjacent to the burner blower housing and connected to the blower housing discharge side by copper tubing. If the blower should lose its pressure, the air safety switch shall close the fuel oil solenoid valve. The air safety switch shall operate satisfactorily (see 4.5.1).

(g) Solenoid valves. Solenoid valves shall withstand HI-shock tests in accordance with type C requirements of MIL-S-901. The solenoid coil shall be constructed for continuous duty and the following:

Voltage rating - 110 ac.
Ambient temperature - 50°C.
Temperature rise - Shall not exceed values specified in MIL-C-2212.
Pressure rating - Shall exceed the fuel oil system relief valve setting.
Enclosure - Spraytight or better.
Terminal block - As required (fitted in the solenoid enclosure).
Locations - Remote.

(h) Fuel oil return control valve. Valve shall be sized to meter fuel oil throughout boiler operation as specified in 3.6.6.2.1.

Pressure rating - as specified (see 6.2.1).

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(i) Combustion safeguard (flame failure safety control). The combustion safeguard may be infrared or ultraviolet. The control shall monitor the pilot and the main flame under all firing conditions and de-energize the fuel oil solenoid valves in 2 to 4 seconds in event of flame failure. The infrared or ultraviolet flame detector shall discriminate between flame and hot refractory or other sources of infrared. The control assembly and the sensing cell or type shall meet HI-shock requirements in accordance with MIL-S-901 and vibration requirements in accordance with MIL-STD-167-1.

(j) Control circuit transformer. The transformer shall have winding voltages, frequency and output ratings as required. The transformer shall conform to commercial marine standards and the following:

Volt ampere rating - 150 percent of total rated load of boiler.
Primary voltage - as specified (see 6.2.1).
Secondary voltage - 110 volts single phase ac.
Enclosure - Spraytight or better.
Voltage regulation - As required.
Duty - Continuous.
Terminal posts - As required.

(k) Ignition transformer. The ignition transformer shall conform to commercial marine standards and the following:

Volt ampere rating - As required.
Primary voltage - 110 volts, single phase.
Secondary voltage - 10,000 $\pm$ 250 volts.
Enclosure - Spraytight or better for remote mounted units.
Mounting location - Local or remote as required.
Ambient temperature - 50°C.
Electromagnetic interference (EMI) suppression - Required (see 3.9).
Duty - Intermittent.

(l) Modulating motor. The modulating motor shall conform to commercial marine standards and the following:

Timing - 15 or 30 seconds as required.
Stroke travel - As required.
Load torque foot-pounds - 150 percent of calculated load.
Voltage - As required by circuit design.
Location - Remote.
Enclosure - Dripproof.

(m) Rotary inverter. The rotary inverter used to supply ac power required for the boiler control and ignition circuit shall be in accordance with MIL-G-3124, MIL-R-2729, and the following:

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Size kW - 150 percent of total rated load of boiler.
Voltage output - 110 volts.
Phase - Single.
Service - Navy A.
Cycle - 60 Hz.
Prime mover - Motor.
Type - Revolving armature.
Enclosure - Dripproof protected.
Speed - Constant.
Duty - Continuous.
Bearings - Ball.
Ambient temperature - 50°C.
Insulation - A, B, or H.
EMI suppression - Required.

(n) Low-water cutoff control. The low-water control shall be of the electrode probe type and for flange mounting on boiler drum of shell. The probes shall be located and arranged to form an isolated circuit, ungrounded. Each probe shall be mounted independently. Probes, complete with cage or surge chamber, shall be removable through companion opening in drum or shell. The electrodes shall be insulated with polytetrafluoroethylene. A ground indicating circuit shall be wired into the cutoff branch of the control circuit. Switches and indicators shall be located on the control panel.

(o) Motor controllers. Electric controllers (motor starters) shall be in accordance with the following:

(1) They shall be mounted remote to the boiler control circuit enclosure.
(2) MIL-C-2212 for ac and dc controllers.
(3) The controllers shall conform to the following classification requirements:

Service - Navy A (HI-shock).
Master switch - Spraytight three elements push button, marked "Start", "Stop", and "Emergency Run".
Ratings - As required.
Enclosure - Dripproof.
Operation - Magnetic.
Type - Resistance for dc units and across line for ac units.
Function - Motor starting.
Duty - Continuous.
Protection - Overload protection with emergency run feature and low voltage protection.
Performance - Automatic or semiautomatic as required.
Ambient temperature - 50°C.

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(p) Thermoplastic material shall not be used.

(q) Console or control enclosure. The enclosure shall be constructed in accordance with class I, dripproof requirements of MIL-E-2036. The enclosure may be a console type or mounted on an operating panel consolidating all indicators, lights and push buttons thereon. The service of each push button indicator and light shall be marked. Particular care shall be taken in design and arrangement of all control equipment to ensure safety of operating personnel from danger of electric shock. Electric motor controllers (starters) shall be arranged for remote mounting. The main control circuit transformer and ignitor transformer shall be located outside the control circuit enclosure.

(r) Indicator lights. Indicator lights shall be provided to indicate main power available, boiler run status, blower motor, fuel pump and feed pump status and all casualties.

(s) Audible alarm. An audible alarm shall be provided with the boiler that will alert the operator when the boiler has been secured after a casualty. This alarm shall be a bell or buzzer with an output of 85 to 105 decibels.

(t) Boiler Master Automatic/Manual Control Station. An automatic/manual (A/M) station shall be provided that will allow the operator to control the air flow/fuel flow to the boiler in automatic or remote manual control from 12 to 120 percent of normal operating pressure (see table I).

3.6.8 Electric motors. Electric motors for driven auxiliaries shall be in accordance with MIL-M-17556, MIL-M-17413, MIL-M-17059 or MIL-M-17060. The motors shall conform to the following classification requirements:

Horsepower - As required.
Volts - As specified (see 6.2.1).
Number of phases - As specified (see 6.2.1).
Frequency - As specified (see 6.2.1).
Speed - Constant.
Duty - Continuous.
Enclosure - Dripproof protected.
Service - Navy A.
Ambient temperature - 50°C.
Insulation - A, B, or H
Type - Induction.
Winding - Squirrel cage for ac motors.
Winding - Shunt (for dc motors).
Bearing - Ball.
Design - B (for ac motors).
EMI suppression - Required (for dc motors).

3.6.8.1 Coupled flexible couplings. Wherever practicable, the drive and driven shall be direct coupled with flexible coupling. When employed, belt drives shall have the following features:

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(a) Matched belts shall be furnished for multiple belt drives.
(b) Belt speeds in excess of 5,500 feet per minute shall not be acceptable.
(c) Drive and driven sheaves shall be in satisfactory static or dynamic balance in accordance with internally excited vibration, type II requirements of MIL-STD-167-1.
(d) Axial movement in drive and driven shall be controlled by thrust bearings. Thrust bearings may be of the combined radial and thrust rolling contact type.

3.7 Operating environment. The boiler construction shall permit normal operation with a momentary roll up to 45 degrees to either side, a permanent list of 15 degrees to either side, or a permanent inclination of 5 degrees fore and aft with the boiler installed with longitudinal axis of the shell assembly in fore and aft direction (see 3.6.2).

3.8 Weight and physical dimensions. The boilers, class I and II, complete with insulation and lagging, all auxiliaries, accessories and controls (with exceptions of motor controllers) shall not exceed the limitations specified in table II.

TABLE II. Physical limitations.

Size	Capacity rated overload (pounds of steam per hour)		Length (ft-in)	Width (ft-in)	Height (ft-in)	Dry weight ac units (lbs)	Dry weight dc units (lbs)
1	500		5-0	3-3	4-6	1800	2000
2	800		6-0	3-6	4-9	2700	3000
3	1200	1400	6-6	3-9	5-0	3000	3400
4	1500	1800	7-0	4-3	6-0	4000	4500
5	2500	3000	8-0	4-9	6-9	6500	7000
6	4000	4800	9-3	5-6	7-3	8500	9000
7	5000	6000	9-6	6-0	8-0	10,000	10,500

3.9 EMI suppression. The combustion and feedwater control system complete, and the circuit component enclosures, wiring conduit, relays switches, transformers and similar equipment comprising the complete circuit shall conform to basic requirements for electric power equipment in accordance with MIL-E-917 and as specified for the individual equipment herein. The combustion control system complete with ignition system shall be constructed to minimize EMI both radiated and conducted, through the use of enclosures, shielded conduits, connectors and line filters. External leads shall be shielded to reduce EMI by shielded metallic conduit of proper diameter with spherical-seal ball and cone fittings in accordance with MIL-S-16165. Electrical connections and wiring shall be waterproof. The unit shall conform to the applicable requirements of MIL-STD-461 for surface ships. When specified in the contract or order, an EMI control plan shall be prepared (see 6.2.2).

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3.10 Identification plates. Each boiler shall have an identification plate conforming to type A or B of MIL-P-15024 and MIL-P-15024/5, except that identification plates of phenolic compounds shall not be acceptable. The identification plates shall contain the following information:

(a) Name and description of article.
(b) Serial.
(c) National stock number.
(d) Name and address of manufacturer.
(e) Contract number.
(f) A square of about 3/4 inch with letter "INSP" at the top for affixing Government inspector stamp.
(g) The words "U.S.".
(h) Design pressure, lb/in^2.
(i) Operating pressure, lb/in^2.
(j) Hydrostatic test pressure, lb/in^2.
(k) Safety valve setting, lb/in^2.
(l) Electrical characteristics.

3.11 Tools. The set of special tools shall be as specified (see 6.2.1 and 6.8). Special tools are defined as those not listed in the Federal Supply Catalog. (This catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)). Each tool shall be indelibly marked to indicate the services for which intended.

3.12 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribe requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. When specified in the contract or order an inspection system program plan shall be prepared (see 6.2.2).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the tests specified in 4.5. When specified in the contract or order, a first article inspection procedure and report shall be prepared (see 6.2.2).

4.4 Quality conformance inspection. Quality conformance inspections shall be conducted on each boiler and shall consist of the tests specified in 4.5. When specified in the contract or order, test reports shall be prepared (see 6.2.2).

4.5 Tests. Tests shall establish conformance to the requirements of this specification and fulfillment of the capacity and efficiency requirements and satisfy control features and safety devices. Safety devices shall be tested by simulating failure conditions requiring action of each safety device.

4.5.1 Operation test. Operation tests shall consist of the tests specified in table III.

TABLE III. Operation tests.

Test	First article	Quality conformance
Steady steaming at 120 percent capacity	8 hours consecutively	1 hour
Steady steaming at rated capacity	24 hours consecutively	1 hour
Steady steaming at 50 percent rated capacity	1 hour	1 hour
Steady steam at minimum capacity	8 hours	1 hour
Steaming at loads modulated from 120 percent capacity to minimum capacity	8 hours	----
Steaming at low loads on-off cycling	300 cycles consecutively	10 cycles consecutively

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TABLE III. Operation tests. - Continued

Test	First article	Quality conformance
Hydrostatic test (see 3.3.5)	1 test	1 test
Safety valve by steam	1 test	1 test
Combustion safeguard (see 3.6.7.1(i))	10 cycles consecutively	5 cycles consecutively
Low water cutoff (see 3.6.2)	10 cycles consecutively	5 cycles consecutively
High stream pressure cutout and air safety switches (see 3.6.7.1(f))	10 cycles consecutively	5 cycles consecutively
Boiler wet/dry weights	Once	----
Low voltage	1 cycle	1 cycle
Chemical and physical properties of oil	1	----
Shock test	1 test	----
Safety valve hand easing gear (if installed) (see 3.3.11 and 4.5.4)	1 test	1 test

4.5.1.1 Operation data. During the operational runs specified in table III, the following data shall be obtained and recorded every 15 minutes, except for the steady steaming run on the first article unit where data shall be obtained and recorded once every hour. Efficiency of the boiler shall be calculated. Performance curves of the boiler shall be plotted. Data are required as follows:

(a) Steam pressure.
(b) Steam quality (see 3.5.2).
(c) Fuel pressure.
(d) Fuel consumption.
(e) Feedwater pump discharge pressure.
(f) Feedwater temperature.
(g) Feedwater input.
(h) Blower discharge pressure.1/
(i) Blower air inlet temperature.1/
(j) Burner plenum pressure.
(k) Blower inlet pressure or barometric pressure (as applicable).1/

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(l) Water level in gauge glass (see 3.6.2).
(m) Gas analysis (carbon dioxide, oxygen, and carbon monoxide).
(n) Boiler casing temperature (see 3.3.16.2).1/
(o) Gas pressure and temperature at rear head.1/
(p) Gas pressure and temperature at breaching outlet.
(r) Motor amperes.
(s) Revolutions per minute of each drive and driven.
(t) Total electric power consumption.

1/ Not required for quality conformance inspection.

4.5.2 Shock test. When specified (see 3.3.4), the first article unit shall pass the grade B shock test in accordance with MIL-S-901.

4.5.3 Hydrostatic test. Before the operational runs, the boiler shall be hydrostatically tested to 150 percent of design pressure to test for strength and tightness (see 3.3.5).

4.5.4 Safety valve hand easing gear. The safety valve hand easing gear shall be tested by fully lifting the safety valve off its seat and the safety valve allowed to blow freely for a short period. This shall be done when the boiler is being brought up to operating pressure at approximately 50 to 75 lb/in^2 (see 3.3.11).

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Boiler and auxiliaries. Unless otherwise specified, the boiler and auxiliaries shall be preserved in accordance with level A, B, or commercial of MIL-B-3180 (see 6.2.1).

5.1.1 Tools. Tools shall be preserved level A, or commercial in accordance with MIL-P-17286, MIL-E-16298, MIL-E-17555, as applicable.

5.1.2 Fiberboard. Unless otherwise specified (see 6.2.1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes shall meet the flame spread index of PPP-F-320.

5.2 Packing. Unless otherwise specified, packing of the boiler and auxiliaries shall be level A, B, or commercial as specified (see 6.2.1).

5.2.1 Level A. Level A packing shall be as specified in 5.2.1.1 and 5.2.1.2.

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5.2.1.1 Boiler. Boilers shall be packed in sheathed crates conforming to MIL-C-104.

5.2.1.2 Auxiliaries. Detached boiler auxiliaries, accessories and controls which are shipped (knocked down) for installation shall be packed in overseas type wood cleated plywood, nailed wood or fiber boxes conforming to PPP-B-601, PPP-B-621 or PPP-B-636 (class 2), respectively, at the option of the contractor. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds. Single items having a gross weight exceeding 200 pounds shall be packed in boxes conforming to PPP-B-601, or PPP-B-621, modified in accordance with the applicable box specification. Boxes shall be lined with a sealed caseliner conforming to MIL-L-10547.

5.2.2 Level B. Level B packing shall be as specified in 5.2.2.1 and 5.2.2.2.

5.2.2.1 Boiler. Boilers shall be packed in sheathed crates conforming to MIL-C-104. Boilers having a gross weight less than 10,000 pounds may be individually packed in unsheathed creates conforming to MIL-C-3774 with the following additional requirements:

(a) The burner or front end of the boiler shall be completely enclosed in a plywood housing. The housing shall be anchored to the skid base of the crate and securely strapped to the boiler.
(b) The entire contents of the crate shall be shrouded approximately two-thirds of the way down with waterproof barrier material conforming to PPP-B-1055. The shrouding shall be securely fastened and arranged in a manner to prevent the formation of water pockets.

5.2.2.2 Auxiliaries. Detached boiler auxiliaries, accessories and controls which are shipped (knocked down) for installation shall be packed as specified in 5.2.1.2. Boxes shall be of the domestic type or class. Caseliners are not required.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2.1), level A and B interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129, commercial interior packs and shipping in accordance with ASTM D 3951.

5.4 Unpacking instructions. Unpacking instructions shall be provided and shall be sealed in waterproof envelopes which shall be tacked, stapled or otherwise securely attached to the outside of each basic equipment container in the most protected location. The words "UNPACKING INSTRUCTIONS" shall be stenciled on each container indicating location of such instructions.

6. NOTES

6.1 Intended use. The fire tube auxiliary boilers covered by this specification are intended for use in shipboard auxiliary steam systems to provide auxiliary steam for galley, laundry, heating, evaporators, steam blanket, and so forth.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, size, and capacity of boiler (see 1.2, 3.8, table I).
(c) When first article inspection is required (see 3.1).
(d) Whether the boiler shall be right or left hand (see 3.3.1).
(e) Type of fuel to be burned (see table I).
(f) Electric power supply available for operation of the boiler and auxiliaries (see 3.3.3 and 3.6.7).
(g) If shock test is required (see 3.3.4).
(h) Type of burner operation for boiler sizes 3 through 7 (see 3.3.6.3.2 and 3.6.1.2).
(i) Pressure rating and primary voltage required for control circuit transformer (see 3.6.7.1).
(j) Voltage, number of phases, and frequency of electric motors (see 3.6.8).
(k) Sets of special tools required (see 3.11).
(l) When rotary inverter supplies power (see 3.3.20(i)).
(m) Level of preservation, packaging and packing required (see 5.1 and 5.2).
(n) When fiberboard boxes are to meet the flame spread index of PPP-F-320 (see 5.1.2).
(o) Special marking required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.9	Electromagnetic interference control plan	DI-EMCS-80199	----
3.12	Drawings, engineering and associated lists	DI-E-7031	----
4.1.2	Inspection system program plan	DI-R-4803	----
4.3	First article inspection procedure	DI-T-4901	----
4.3	First article inspection report	DI-T-4902	----
4.4	Reports, test	DI-T-2072	10.1.b

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.2.2 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be listed on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Capacity. Capacity of the boiler is defined as pounds of saturated steam per hour (actual evaporation) at the specified operating pressure with feed water and air temperature at 100°F.

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6.7 Operating pressure. Operating pressure of boilers equipped for automatic burner operation is approximately the mean between the automatic burner cut-in and cutoff pressures.

6.8 Special tools. The set of special tools should include the following:

1 - Tube expander and mandrel for each size of tube used.
1 - Manhole cover wrench, if required.
1 - Handhole cover wrench.
1 - Ripping chisel.
1 - Oyster knife.
1 - Set of tube plugs (blind nipple type of use with expander and mandrels furnished for tubes) consisting of 10 percent of the number of each size of tube installed in the boiler.
1 - Set of safety valve gags.
1 - Safety valve seat and disc templates.
1 - Set of gasket templates (one for manhole, if required, and one for each size of handhole). The template shall be of steel 1/16-inch thick.
2 - Brushes, wire-bristle, fire tube cleaning.
1 - Brush handle, sectionalized, for fire tube cleaning brush.
1 - Manual ignition torch with torch holder.
1 - Set of feedwater pump tools and wrenches as required.
1 - Set of tool boxes (metal) as required.

6.9 Subject term (key word) listing.

Feedwater
Hand easing gear
Purging
Steam blanket
Surface blow pipe
Water gauges

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4410-N056)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-16907G(SH)	BOILERS, STEAM, HIGH PRESSURE, AUXILIARY FIRE TUBE TYPE NAVAL SHIPBOARD

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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